Binary Number System

10101010101010101010101010101010
01010101010101010101010101010101
10101010101010101010101010101010

“Normal” Number System

- Decathlon = 10 events
- The **Dec**imal system that is known as our conventional number system uses a BASE of 10.
- There are 10 digits that we can use with this system…

0 1 2 3 4 5 6 7 8 9

Number Places (Decimal)

100000s	**10000s**	**1000s**	**100s**	**TENS**	**ONES**
10^5 =100000	10^4 =10000	10^3 =1000	10^2 =100	10^1 =10	10^0 =1
			2	5	7

Example number: 257

Can be represented as 2 x 100 + 5 x 10 + 7 x 1 = 200 + 50 + 7 = 257

Binary Number System

- Bi-cycle = 2 wheels
- The Binary number system uses a BASE of 2
- There are two digits that we can use…
- 1 0
- In computers, transistors are little switches that are either ON (1) or OFF (0)

Number Places (Binary)

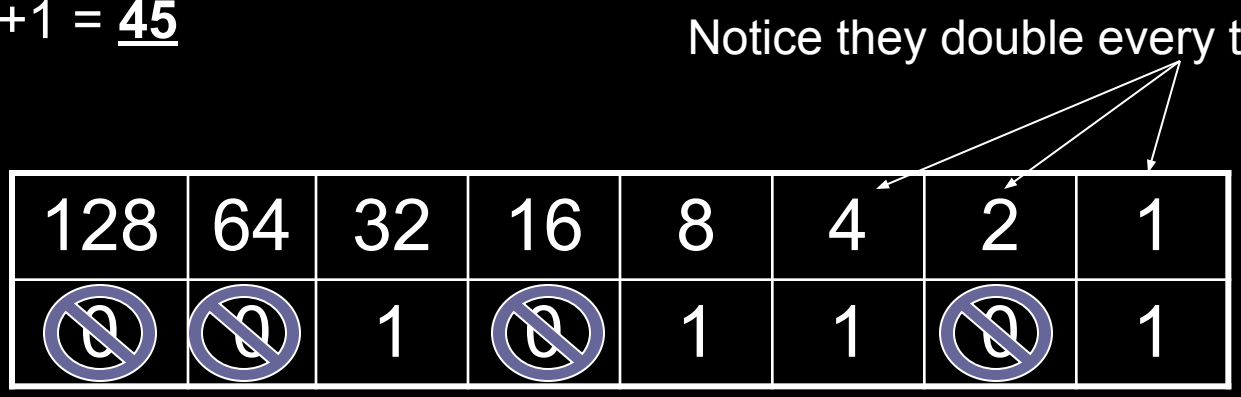
2^7 = 128	2^6 = 64	2^5 =32	2^4 =16	2^3 =8	2^2 =4	2^1 =2	2^0 =1
				1	0	1	1

1 x 8 = **8** | 0 x 4 = **0** | 1 x 2 = **2** | 1 x 1 = **1**

Example number 1011.
= 1 x 8 + 0 x 4 + 1 x 2 + 1 x 1
= 8 + 0 + 2 + 1
=11
Therefore, the number 1011 in Binary is equal to 11 (in decimal).

For the first few conversions… (Binary to Decimal)

- Do the following steps:
  - Draw the little table
  - Write in the binary number (Example: 101101)
  - Add them up:
    - 32 + 8 + 4 +1 = **45**



Lets Try Another One!

- Binary number 100110 = ?

128	64	32	16	8	4	2	1
		32	0	0	4	2	0

- Now do some on your own!

Convert the following binary numbers to equivalent decimal numbers.

(a) $(1101)_2$ 13

(b) $(11101)_2$ 29

(c) $(0101\ 1101)_2$ 93

(d) $(1101\ 1101)_2$ 221

(e) $(1111\ 1111)_2$ 255

(f) $(0101\ 1001)_2$ 89

(g) $(1101\ 1101\ 0101)_2$ 3641

(h) $(11100.101)_2$ 28.625

Converting from Decimal to Binary

- We start on the **left** this time and try to “take away” the largest number we can
- We use the same table:

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1

- Example: **15**
  - 128, 64, 32, and 16 are all too big!
  - So the largest number we can use is 8
  - Now 15 – 8 = 7 so we now repeat the steps using 7
  - 7-4 = 3 repeat using 3
  - 3-2 = 1
  - THEREFORE, 15 (in decimal) is equal to 1111 in Binary

Let's try another one

- Convert the Decimal Number 75 to Binary

128	64	32	16	8	4	2	1

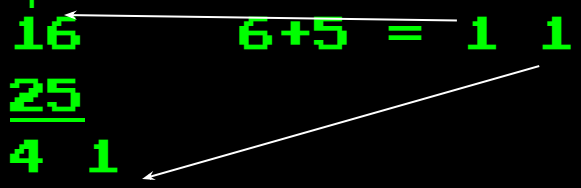
- Now do some on your own!

Convert the following decimal numbers to equivalent binary numbers.

- (a) $(57)_{10}$ 111001_2
- (c) $(255)_{10}$ 11001_2
- (d) $(256)_{10}$ 100000000_2
- (e) $(2416)_{10}$ 100101110000_2
- (f) $(4195)_{10}$ 1000001101011_2

Adding Decimal Numbers

- When adding decimal numbers (base 10) we carry over when we get to "10 or greater".
- See:



Now Adding Binary

- Using the same logic, in Binary we will carry a digit when we get to ____.
- Let's do some examples:

Now Adding Binary

- Using the same logic, in Binary we will carry a digit when we get to ____.
- Let's do some examples:

```
    1 1
    1 0 1      5
+     1 1      3
  1 0 0 0      8
```

Now Adding Binary

- Using the same logic, in Binary we will carry a digit when we get to ____.
- Let's do some examples:

1 1	
1 0 1 0	10
1 1 0	6
10 0 0 0	16

Now Adding Binary

- Using the same logic, in Binary we will carry a digit when we get to ____.
- Let's do some examples:

1 1 1 1	
1 1 0 1 1	27
1 1 1 1	15
101 0 1 0	42

Perform the following unsigned binary arithmetic. Verify your answer by converting each problem into decimal. (Note: the last two are subtraction!)

a. 0111 0101
+ 0011 0011
1010 1000

b. 0010 0110
+ 0101 1011
1000 0001

c. 1001 0011
+ 0011 1011
1100 1110

d. 0101 1100
+ 0001 1111
0111 1011

More Practice

Play the following binary decimal conversion game.

https://learningcontent.cisco.com/games/binary/index.html